MINISTRY OF
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INFORMATION TECHNOLOGY

ISEA

DSCI
PROMOTING DATA PROTECTION

FORENSICS LOG REPORT

YEAR 2025-2026

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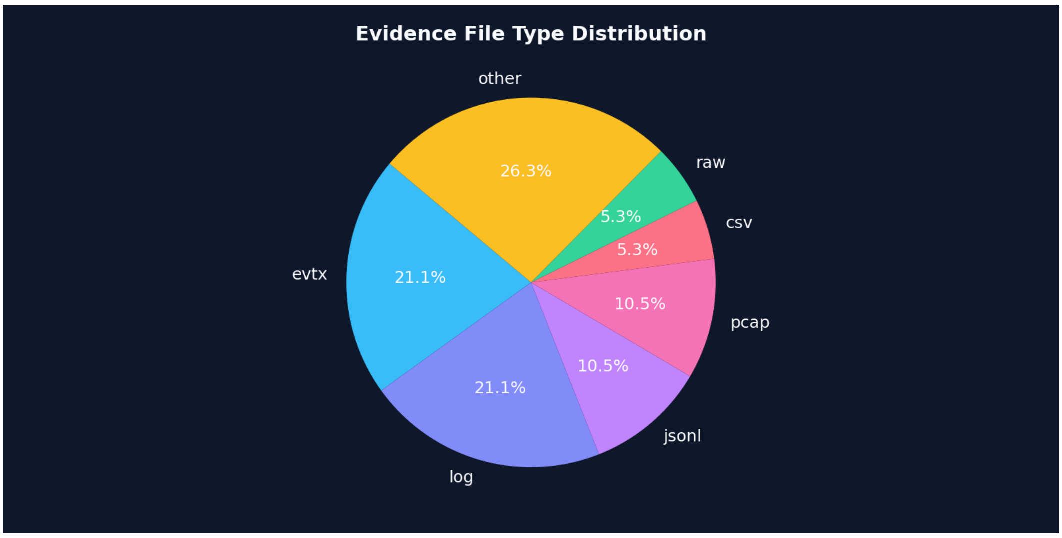
1. Case Snapshot and Dataset Overview

Case CASE-2990-X42 was processed to summarize evidence intake, integrity, custody continuity, normalization quality, behavioral signals, and anomaly outcomes.

The dataset contains 19 artifacts with a combined size of 547.87 MB. The observed upload window spans from 2026-02-09 01:12:11 to 2026-02-09 04:05:27.

Most frequent formats in this dataset: evtx(4), log(4), jsonl(2), pcap(2), csv(1). These distributions help anticipate which parsers and correlation paths will contribute most to the investigation.



2. Evidence Intake Summary (Files, Formats, Sizes, Upload Times)

Evidence intake summarizes what was received, when it was received, and how it was categorized for downstream processing.

Artifacts are grouped by format and size to identify heavy sources (for example, memory dumps or PCAP segments) and to verify that the expected telemetry types are present.

The table below highlights the largest artifacts, which typically dominate processing time and often hold high-value forensic context.

Largest Artifacts (Top 10)

Filename	Type	Size	Upload Time
memory_dump_host07.raw	raw	512.00 MB	2026-02-09 03:10:44
pcap_segment_20260209_002.pcap	pcap	7.00 MB	2026-02-09 03:22:07
pcap_segment_20260209_001.pcap	pcap	6.00 MB	2026-02-09 03:18:22
audit_export_20260209.zip	zip	4.00 MB	2026-02-09 03:25:55
edr_process_events_20260209.jsonl	jsonl	3.00 MB	2026-02-09 02:24:10
sysmon_20260209_0001.evtx	evtx	2.34 MB	2026-02-09 01:19:22
sysmon_20260209_0002.evtx	evtx	2.20 MB	2026-02-09 01:21:40
firewall_traffic_20260209.csv	csv	2.00 MB	2026-02-09 02:18:06
win_security_20260209_0001.evtx	evtx	1.76 MB	2026-02-09 01:12:11
win_security_20260209_0002.evtx	evtx	1.64 MB	2026-02-09 01:13:58

3. Integrity Status (Per-File Hash Verification Results)

Per-file integrity verification ensures evidence remains bit-consistent from intake through processing.

Overall distribution: VALID=18, CORRUPTED/FAIL=1.

Any corrupted or failed artifacts should be isolated for re-acquisition or manual handling to avoid contaminating downstream analytics.

Integrity Status (First 20)

Filename	SHA-256 (short)	Status
win_security_20260209_0001.evtx	a3b9c1d5e7f2…	VALID
win_security_20260209_0002.evtx	b4c2d1e0f9a8…	VALID
sysmon_20260209_0001.evtx	c1d2e3f4a5b6…	VALID
sysmon_20260209_0002.evtx	d0e1f2a3b4c5…	VALID
nginx_access_20260209_01.log	e5f4d3c2b1a0…	VALID
nginx_access_20260209_02.log	f1a2b3c4d5e6…	VALID
nginx_error_20260209.log	0a1b2c3d4e5f…	VALID
firewall_traffic_20260209.csv	1b2c3d4e5f60…	VALID
dns_queries_20260209.jsonl	2c3d4e5f6071…	VALID
edr_process_events_20260209.jsonl	3d4e5f607182…	VALID
memory_dump_host07.raw	4e5f60718293…	VALID
pcap_segment_20260209_001.pcap	5f60718293a4…	VALID
pcap_segment_20260209_002.pcap	60718293a4b5…	VALID
audit_export_20260209.zip	718293a4b5c6…	VALID
host_inventory_20260209.xlsx	8293a4b5c6d7…	VALID
suspicious_script_01.ps1	93a4b5c6d7e8…	CORRUPTED
webshell_suspect.aspx	a4b5c6d7e8f9…	VALID
linux_auth_20260209.log	b5c6d7e8f90a…	VALID
cloudtrail_20260209.json	c6d7e8f90a1b…	VALID

4. Chain-of-Custody Status (Hash Chain Verification Results)

Chain-of-custody status summarizes whether custody continuity was preserved across the evidence lifecycle.

Hash chain status: VALID. Breaks observed: 0. First break: N/A.

If breaks are present, evidence after the first break should be treated as disputed until independently validated.

Hash Chain Summary

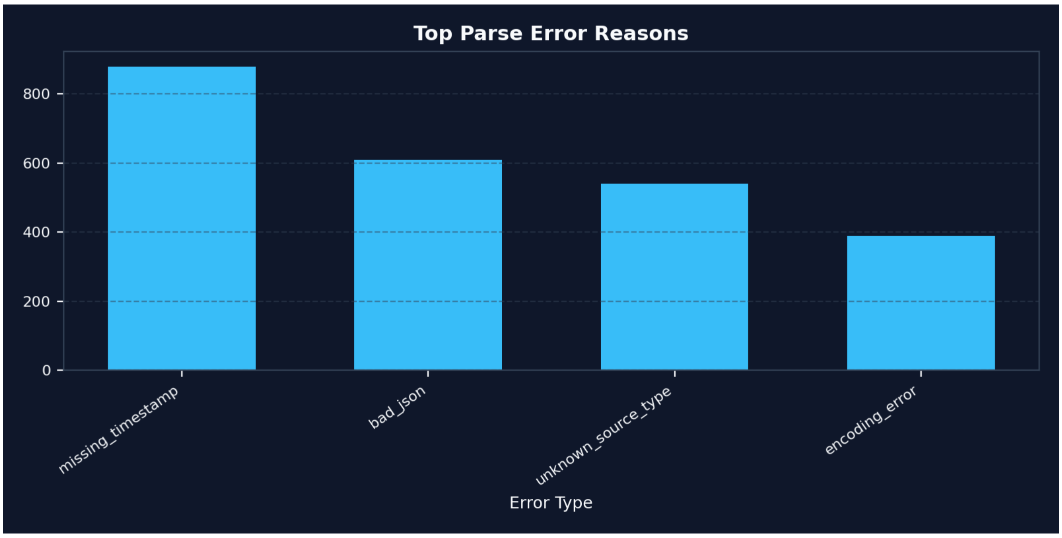
Status	Breaks	First Break
VALID	0	N/A

5. Parsing and Normalization Summary (Parsed vs Unparsed Records)

Parsing and normalization summarize how much raw telemetry could be converted into structured records ready for correlation.

Records: total=152400, parsed=149980, unparsed=2420.

Top parse error reasons: missing_timestamp=880, bad_json=610, unknown_source_type=540, encoding_error=390. Addressing dominant error types typically improves coverage and reduces blind spots.



6. Hot Store Ingestion Summary (DuckDB Tables and Row Counts)

Hot store ingestion summarizes which normalized datasets were materialized for fast querying and analytics.

DuckDB table counts are a quick sanity check: they confirm that event streams, entities, indicators, and alerts were actually persisted after parsing.

If any expected table is missing or near-empty, it usually indicates an upstream parsing or mapping issue.

DuckDB Materialization

Table	Rows
raw_events	152400
normalized_events	149980
entities	4870
ioc_indicators	412
alerts	186
timeline	96

7. Behavioral Feature Summary (Signals Generated and Coverage)

Behavioral features describe what signals were computed from normalized records (for example: login velocity spikes, impossible travel, privilege escalations).

Signal counts give a quick view of coverage: higher counts often represent broad, low-confidence indicators; smaller counts can represent high-precision detections.

The chart below highlights the most frequent signals in this dataset.

Data Coverage

Expected Field	Present
signals	Yes

8. Anomaly Detection Summary (Total Scored, Anomalies Flagged, Score Range)

Anomaly detection summarizes how many records were scored and how many were flagged above the anomaly threshold.

Scored=149980, flagged=186, score range=0.03 to 0.98.

A tight score range usually indicates stable behavior; sharp peaks or long tails typically indicate bursty or staged activity.

Data Coverage

Expected Field	Present
anomaly_detection	Yes
scored_records	Yes
anomalies_flagged	Yes
score_min	Yes
score_max	Yes
scores	Yes

9. High-Risk Entities and Alerts (Users/IPs, Risk Levels, Key Triggers)

High-risk entities consolidate alerts by principal (user, host, IP, domain) and provide a ranked view of the most concerning actors.

A small number of entities frequently account for most high-confidence risk, making them good candidates for triage and containment checks.

The table below lists the highest-risk items and the triggers that caused scoring.

Top Alerts (First 15)

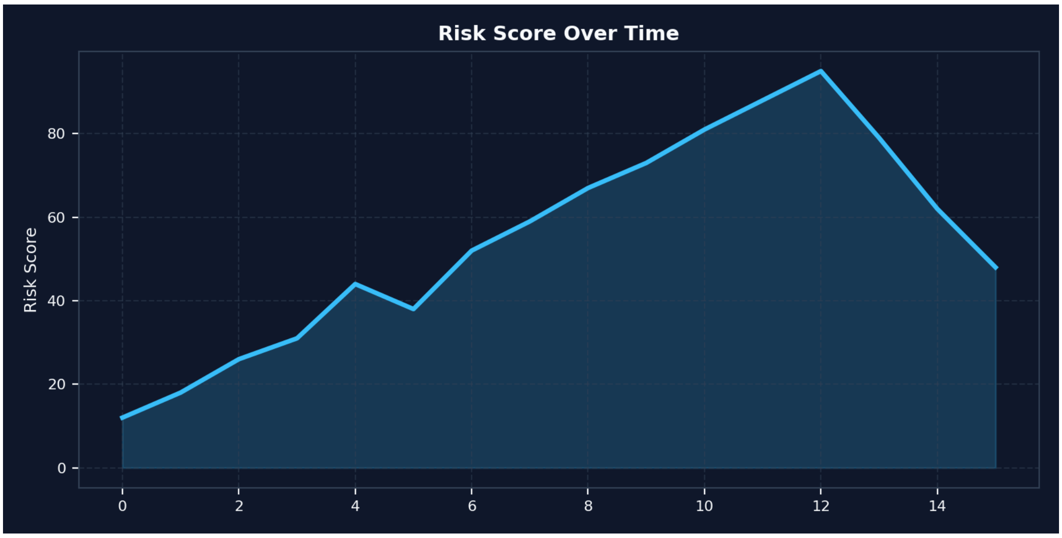
Entity	Risk Score	Risk Level	Key Triggers
user:svc_backup	96	CRITICAL	password_spray, privilege_escalation, log_tamper
ip:185.231.72.19	93	HIGH	suspicious_outbound, dns_dga_pattern
user:admin02	91	HIGH	login_velocity_spike, impossible_travel
host:APP-SRV-07	89	HIGH	rare_process_spawn, persistence_attempt
domain:cdn-checks[.]com	86	HIGH	dns_dga_pattern, suspicious_outbound
user:hr01	82	MEDIUM	geo_jitter, login_velocity_spike
ip:45.9.148.201	84	MEDIUM	suspicious_outbound
host:DB-SRV-02	80	MEDIUM	large_data_transfer, suspicious_outbound

10. Investigation Timeline Highlights and Recommended Actions

Timeline highlights summarize how risk evolved over the observed window and identify the most important time slices for deeper artifact correlation.

Peak windows are the best starting point for reconstruction: pivot from the timestamp into supporting logs, process events, DNS, and network captures.

Recommended actions are derived from the dominant triggers observed in signals and entity alerts.



Peak Windows (Top 5)

- 2026-02-09 03:27:44 — risk 95 — Peak anomaly wave; high-confidence suspicious activity window.
- 2026-02-09 03:12:55 — risk 88 — Multiple synchronized high-risk triggers correlate to same window.
- 2026-02-09 03:01:42 — risk 81 — Log tamper signals detected (sudden gaps / truncation behavior).
- 2026-02-09 03:42:10 — risk 79 — Outbound connections reduced after containment-like behavior.
- 2026-02-09 02:45:10 — risk 73 — Large data transfer spikes from DB-SRV-02 to external endpoint.

Recommended Actions

- Enforce password reset for affected accounts; enable MFA for all privileged users; review lockout policy.
- Audit recent privilege changes and group membership updates; rotate admin credentials; validate sudo/admin use.
- Block suspicious domains at resolver and perimeter; check endpoints for beaconing; review DNS logs for related subdomains.
- Inspect outbound traffic for data exfil paths; quarantine endpoints showing repeated high-volume transfers; validate destinations.
- Review scheduled tasks/services/startup entries; inspect script execution history; collect EDR triage for impacted hosts.
- Preserve original logs; verify collector integrity; compare against upstream sources; expand timeline window for gaps.
- Document all actions with timestamps and operator identity to preserve auditability.